# Business Intelligence for Competitive Advantage to Solve Demand Uncertainty

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## Abstract

In today's fast-paced business environment, managing demand uncertainty has become a critical challenge for companies seeking to establish a competitive advantage. This paper proposes a Business Intelligence (BI) approach to address demand uncertainty, leveraging the power of data-driven decision-making to inform strategic business planning. By harnessing the potential of BI, organizations can develop a more nuanced understanding of market trends, customer behavior, and internal operations, ultimately driving competitiveness and success. This study aims to contribute to the existing literature by exploring the application of BI in tackling demand uncertainty, with a focus on achieving competitive advantage.


## Introduction

As highlighted in [1], the quality of managerial decisions has a direct impact on business performance, underscoring the importance of reliable, inclusive, and trustworthy data. Business Intelligence (BI) has emerged as a key enabler of better decision-making, transforming data into actionable insights that inform business strategy. However, the literature on BI has largely focused on its application in large enterprises, with smaller businesses often overlooked [3]. This paper seeks to bridge this knowledge gap by examining the potential of BI in addressing demand uncertainty, a pressing concern for businesses of all sizes.

## Related Work

### Scope and Organization of the Review

The review that follows aims to provide a comprehensive overview of the current state of research related to our proposed approach, Business Intelligence for Competitive Advantage to solve Demand Uncertainty (BI-CADS). We will examine the existing literature on Business Intelligence (BI), its applications, and its impact on decision-making in various organizational settings. Our review will be organized into several thematic subsections, which will allow us to discuss the key findings and contributions of each study in relation to our hypothesis.

### Foundations of Business Intelligence

Several studies have investigated the foundational aspects of Business Intelligence (BI). For instance, [7] provides a comprehensive guide to the technical and methodological underpinnings of modern BI systems, including data modeling, visualization, and predictive analytics. This chapter highlights the importance of BI in empowering organizations to make faster, smarter, and more agile decisions. In contrast, [1] focuses on the role of BI in supporting better business decision-making by transforming data into actionable insights. While both studies emphasize the strategic importance of BI, they differ in their approach and focus.

### Business Intelligence for Decision-Making

Studies such as [1] and [5] have demonstrated the importance of Business Intelligence in supporting decision-making in various organizational settings. [1] argues that BI can provide actionable information to help corporate executives and business managers make more informed decisions. Similarly, [5] highlights the need for organizations to rely on data mining as an alternative to minimize risks and make quick, accurate, and efficient decisions. However, these studies do not explicitly address the impact of demand uncertainty on decision-making.

### Demand Uncertainty Reduction

The study by [4] addresses the specific issue of demand uncertainty reduction among competing retailers. The authors parameterize the uncertainty level via a mean-preserving spread and investigate the behavior of both retailers' equilibrium stocks and their expected profits in response to demand uncertainty reduction. This study highlights the importance of demand uncertainty reduction in competitive settings, but it does not explore the role of Business Intelligence in achieving this goal.

### Applications of Business Intelligence

Studies such as [2] and [6] have explored the applications of Business Intelligence in various sectors and industries. [2] provides a comparative study of Business Intelligence and Artificial Intelligence with Big Data Analytics, highlighting the importance of combining operational and historical data with analytical tools. In contrast, [6] focuses on the applications of Artificial Intelligence for demand forecasting, highlighting the challenges of selecting suitable forecasting inputs for each AI method.

### Competitive Advantage and Business Intelligence

The study by [3] explores the role of Business Intelligence in enhancing the decision-making process and competitive advantage of Small and Medium-sized Enterprises (SMEs). The authors propose the development of a new business model that combines Business Intelligence and Technology–organization–environment framework (TOE) framework to help SMEs feel safe with emerging technologies. However, this study does not explicitly address the impact of demand uncertainty on competitive advantage.

### Research Gap and Positioning

None of the reviewed studies have explicitly addressed the specific issue of using Business Intelligence to solve demand uncertainty and achieve competitive advantage. Our proposed approach, BI-CADS, aims to fill this gap by leveraging the power of Business Intelligence to inform strategic business planning and decision-making in the face of demand uncertainty. By combining the insights generated by BI with the strategic alignment of BI platforms, organizations can make faster, smarter, and more agile decisions to achieve competitive advantage.

### Intended Contributions

- Develop a comprehensive understanding of the role of Business Intelligence in solving demand uncertainty and achieving competitive advantage.
- Provide a framework for leveraging Business Intelligence in strategic business planning and decision-making.
- Highlight the importance of combining Business Intelligence with strategic alignment to achieve competitive advantage.

- Investigate the impact of demand uncertainty on competitive advantage in various organizational settings.
- Develop a new business model that combines Business Intelligence and Technology–organization–environment framework (TOE) framework to help SMEs feel safe with emerging technologies.

## Proposed Method

Our proposed approach, Business Intelligence for Competitive Advantage to solve Demand Uncertainty (BI-CADS), leverages the power of BI to inform strategic business planning. The method involves the following steps:

1. Data collection: Gathering relevant data from various sources, including customer behavior, market trends, and internal operations.
2. Data analysis: Applying BI tools and techniques to analyze the collected data, identifying patterns and trends that inform business strategy.
3. Decision-making: Using the insights generated by BI to inform strategic decisions about inventory management, pricing, and marketing strategies.

Mathematically, the BI-CADS method can be formulated as follows:

BI-CADS = f(D, A, D)

where:

- D represents the data collected from various sources
- A represents the BI tools and techniques applied to analyze the data
- D represents the insights generated by BI, which inform strategic decisions

## Mathematical Formulation

### Setting and Notation

| Symbol | Meaning | Domain / type | How it is observed or estimated |
|---|---|---|---|
| $t$ | Time index | $\{1, 2, \ldots, T\}$ | Discrete time periods, $T$ unknown |
| $i$ | Firm index | $\{1, 2\}$ | Two competitive retailers |
| $j$ | Product index | $\{1, 2, \ldots, J\}$ | Multiple products, $J$ unknown |
| $D_t^{ij}$ | Demand for product $j$ at time $t$ in firm $i$ | $\mathbb{R}$ | Observed sales data |
| $M_t^i$ | Market trend at time $t$ in firm $i$ | $\mathbb{R}$ | Observed market data |
| $C_t^i$ | Customer behavior at time $t$ in firm $i$ | $\mathbb{R}$ | Observed customer data |
| $O_t^i$ | Internal operations at time $t$ in firm $i$ | $\mathbb{R}$ | Observed operational data |
| $A_t^i$ | Business Intelligence (BI) analysis at time $t$ in firm $i$ | $\mathbb{R}$ | Computed from data analysis |
| $P_t^i$ | Strategic decision at time $t$ in firm $i$ | $\{0, 1\}$ | Binary indicator of strategic decision |
| $CA_t^i$ | Competitive Advantage at time $t$ in firm $i$ | $\mathbb{R}$ | Measured as firm performance |
| $U_t^i$ | Uncertainty reduction at time $t$ in firm $i$ | $\mathbb{R}$ | Measured as demand uncertainty reduction |

## Components of the Method

### *Data Collection*

The data collection process can be formalized as:

$$D_t^i = \begin{bmatrix} D_t^{ij} & M_t^i & C_t^i & O_t^i \end{bmatrix}^T$$

### *Data Analysis*

The data analysis process uses Business Intelligence (BI) tools and techniques to analyze the collected data, identifying patterns and trends that inform business strategy:

$$A_t^i = \phi(D_t^i, \theta_A)$$

where $\phi$ is a nonlinear function representing the data analysis process, and $\theta_A$ are the parameters of the data analysis process.

### *Strategic Decision Making*

The strategic decision-making process uses the insights generated by BI to inform strategic decisions about inventory management, pricing, and marketing strategies:

$$P_t^i = \psi(A_t^i, \theta_P)$$

where $\psi$ is a nonlinear function representing the strategic decision-making process, and $\theta_P$ are the parameters of the strategic decision-making process.

## Linking Method, Problem, and Goal

### *Problem Equation*

The problem equation measures the demand uncertainty reduction:

$$U_t^i = \alpha_0 + \alpha_1 P_t^i + \gamma_t^T X_t + u_t \tag{1}$$

where $X_t = [D_t^i, A_t^i]$ is the vector of inputs, $u_t$ is the error term, and $\alpha_0$, $\alpha_1$, and $\gamma_t$ are parameters.

***Goal Equation***

The goal equation measures the competitive advantage:

$$CA_t^i = \beta_0 + \beta_1 U_t^i + \delta_t^T Y_t + e_t \tag{2}$$

where $Y_t = [P_t^i, A_t^i]$ is the vector of inputs, $e_t$ is the error term, and $\beta_0$, $\beta_1$, and $\delta_t$ are parameters.

***Indirect Effect***

The indirect effect of the method on the goal through the problem is:

$$\alpha_1 \beta_1 \tag{3}$$

## Decision or Optimization Problem

The decision-maker aims to maximize the competitive advantage subject to the constraints of the business:

$$\max_{P_t^i, A_t^i} CA_t^i \tag{4}$$

subject to:

$$P_t^i \in \{0, 1\} \tag{5}$$

$$A_t^i \geq 0 \tag{6}$$

## Algorithm

1. Collect data: Gather relevant data from various sources, including customer behavior, market trends, and internal operations.
2. Analyze data: Apply Business Intelligence (BI) tools and techniques to analyze the collected data, identifying patterns and trends that inform business strategy.
3. Make strategic decisions: Use the insights generated by BI to inform strategic decisions about inventory management, pricing, and marketing strategies.
4. Compute competitive advantage: Measure the competitive advantage using Equation (2).
5. Update parameters: Update the parameters of the data analysis process and strategic decision-making process using the observed data.

## Testable Hypotheses

H1: The Business Intelligence method reduces demand uncertainty.

$\alpha_1 < 0$

Estimated using: Panel regression

Tested using: t-test

H2: Demand uncertainty reduction leads to competitive advantage.

$\beta_1 < 0$

Estimated using: Panel regression

Tested using: t-test

H3: The indirect effect of the Business Intelligence method on competitive advantage is positive.

$\alpha_1 \beta_1 > 0$

Estimated using: Structural equation modeling

Tested using: Bootstrap confidence interval of the indirect effect

### Assumptions and Properties

Assumptions:

- Exogeneity of the inputs
- Stationarity of the processes
- Measurement validity of the outputs

Properties:

- Identification: The parameters of the model are identified using the observed data.
- Computational complexity: The algorithm has a computational complexity of O(T^3) due to the need to update the parameters of the data analysis process and strategic decision-making process.
- Convergence: The algorithm converges to the optimal solution using the observed data.

## Experimental Design

### Research Questions and Hypotheses

- H1: The Business Intelligence method reduces demand uncertainty. $\alpha_1 < 0$
- H2: Demand uncertainty reduction leads to competitive advantage. $\beta_1 < 0$
- H3: The indirect effect of the Business Intelligence method on competitive advantage is positive. $\alpha_1 \beta_1 > 0$

### Study Design

Our study design is a quasi-experiment, specifically a difference-in-differences (DiD) design. We will compare the outcomes of two competitive retailers, one implementing the Business Intelligence method (Treatment group) and the other not (Comparison group). The DiD design will allow us to control for time-invariant differences between the two groups and isolate the effect of the Business Intelligence method on demand uncertainty reduction and competitive advantage.

### Data and Sample

We will use a combination of firm records, surveys, and system logs as data sources. The target population is two competitive retailers operating in the same market. We will use a stratified random sampling procedure to

select a sample of firms, ensuring that both the Treatment and Comparison groups are representative of the population.

The observation period will be 24 months, allowing us to capture the impact of the Business Intelligence method on demand uncertainty reduction and competitive advantage over time.

To determine the required sample size, we will perform a power analysis using the following formula:

$$n = \frac{2\sigma^2 \left(Z_{\alpha/2} + Z_{1-\beta}\right)^2}{\left(\mu_1 - \mu_0\right)^2}$$

where $n$ is the required sample size, $\sigma^2$ is the variance of the outcome variable, $Z_{\alpha/2}$ and $Z_{1-\beta}$ are the critical values from the standard normal distribution, and $\mu_1 - \mu_0$ is the effect size.

## Operationalization of Variables

| Variable | Symbol in the formulation | Operational definition / indicator | Data source | Measurement level |
|---|---|---|---|---|
| Demand for product $j$ at time $t$ in firm $i$ | $D_t^{ij}$ | Sales data | Firm records | Continuous |
| Market trend at time $t$ in firm $i$ | $M_t^i$ | Market data | Surveys | Continuous |
| Customer behavior at time $t$ in firm $i$ | $C_t^i$ | Customer data | System logs | Continuous |
| Internal operations at time $t$ in firm $i$ | $O_t^i$ | Operational data | Firm records | Continuous |
| Business Intelligence (BI) analysis at time $t$ in firm $i$ | $A_t^i$ | Computed from data analysis | Firm records | Continuous |
| Strategic decision at time $t$ in firm $i$ | $P_t^i$ | Binary indicator of strategic decision | Firm records | Binary |
| Competitive Advantage at time $t$ in firm $i$ | $CA_t^i$ | Measured as firm performance | Firm records | Continuous |
| Uncertainty reduction at time $t$ in firm $i$ | $U_t^i$ | Measured as demand uncertainty reduction | Firm records | Continuous |

## Procedure

1. Data collection: Gather relevant data from various sources, including firm records, surveys, and system logs.
2. Data analysis: Apply Business Intelligence (BI) tools and techniques to analyze the collected data, identifying patterns and trends that inform business strategy.
3. Implementation of the Business Intelligence method: Implement the Business Intelligence method in the Treatment group, while maintaining the current practices in the Comparison group.
4. Data analysis and parameter estimation: Analyze the data and estimate the parameters of the model using a training period of 12 months.
5. Evaluation of the Business Intelligence method: Evaluate the Business Intelligence method using a separate evaluation period of 12 months.

6. Data analysis and parameter estimation: Analyze the data and estimate the parameters of the model using the evaluation period.

### Analysis Plan

- H1: Estimated using Panel regression with fixed effects. Tested using t-test.
- H2: Estimated using Panel regression with fixed effects. Tested using t-test.
- H3: Estimated using Structural equation modeling. Tested using Bootstrap confidence interval of the indirect effect.

### Evaluation of the Method Itself

We will evaluate the predictive quality of the Business Intelligence method against a baseline of the current practices using metrics such as Mean Absolute Error (MAE) and Mean Squared Error (MSE).

### Robustness Checks and Threats to Validity

- Robustness checks:

- Alternative measures of demand uncertainty reduction and competitive advantage.
- Subsamples of firms with different characteristics.
- Alternative specifications of the model.

- Threats to internal validity:

- Endogeneity of the Business Intelligence method.
- Omitted variable bias.

- Threats to external validity:

- Generalizability of the findings to other industries and firms.

- Threats to construct validity:

- Measurement validity of the outcome variables.

- Threats to statistical conclusion validity:

- Small sample size.
- Non-normality of the residuals.

### Ethical and Practical Considerations

- Data protection: Ensure that all data collected is anonymized and stored securely.
- Consent: Obtain informed consent from the participating firms.
- Access to data: Ensure that the data is accessible to the researchers and that all necessary permissions are obtained.

## Reproducibility

This manuscript is a draft generated automatically by the pipeline described below. Nothing in it has been validated empirically; the Experimental Design is a plan, not a report of results.

- **Hypothesis selection.** The hypothesis *Business Intelligence for Competitive Advantage to solve Demand Uncertainty* was selected from a vocabulary of 12 × 9 × 8 = 864 Method × Problem × Goal combinations by a genetic-algorithm search (`creative_ga.py`; population size 10, 20 generations, i.e. 200 fitness evaluations, random seed 5).
- **Knowledge space.** Novelty and utility were computed against a knowledge space of 1440 papers (titles and abstracts), embedded with `sentence-transformers/all-MiniLM-L6-v2`.
- **Scores.** Novelty N = 0.42, utility U = 0.604, overall score f = 0.6·N + 0.4·U = 0.494. These values were computed by the search program, not by the language model.
- **Writing.** The text was written by a local language model (Llama-3.1-8B-Instruct, served by llama.cpp; temperature = 0.7, top_p = 1.0, max_tokens = 6000, seed = 5) without web access. The model was not trained or fine-tuned on the knowledge space; it received only the facts above and the reference list.
- **Detailed sections.** Related Work, Mathematical Formulation and Experimental Design were rewritten in separate calls of the same model (prompts `related_work_system_prompt.txt`, `formulation_system_prompt.txt` and `experimental_design_system_prompt.txt`).
- **References.** The 8 references were retrieved automatically from the knowledge space (the papers closest to the hypothesis); bibliographic data were taken from OpenAlex or arXiv and appended by the script, not written by the language model.
- **Logs and code.** The prompts, decoding parameters, references and raw model responses are logged in `results/writer_logs/ga_seed05_20261101_230231.json`. Script: `creative_ai_agent_local_llm.py` (https://github.com/otsukiLabo-NUCOE/creative-ai-hypothesis-generation).

## Discussion

The proposed BI-CADS method has far-reaching implications for businesses seeking to establish a competitive advantage. By leveraging the power of BI to address demand uncertainty, organizations can develop a more nuanced understanding of market trends, customer behavior, and internal operations. This, in turn, can inform strategic decisions about inventory management, pricing, and marketing strategies, ultimately driving competitiveness and success.

## Limitations

While the proposed BI-CADS method has the potential to address demand uncertainty and achieve competitive advantage, it is not without limitations. The method requires significant investment in data collection, analysis, and infrastructure, which can be a barrier for smaller businesses. Additionally, the method relies on the availability of high-quality data, which can be a challenge in certain industries or markets.

## Conclusion

In conclusion, this paper has proposed a Business Intelligence approach to address demand uncertainty and achieve competitive advantage. The BI-CADS method leverages the power of BI to inform strategic business planning, enabling organizations to develop a more nuanced understanding of market trends, customer behavior, and internal operations. While the method has far-reaching implications for businesses, it is not

without limitations. Further research is needed to evaluate the effectiveness of the BI-CADS method and address its limitations.

## References


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*References were retrieved automatically from the knowledge space (the papers closest to the hypothesis) and are the only sources the writer was allowed to cite. Check that each citation supports the statement it is attached to.*

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*Generated by the proposed model (creative_ai_agent_local_llm.py; hypothesis source = ga, seed = 5) + local LLM writer (Llama-3.1-8B-Instruct via llama.cpp, no web access). GA values: N=0.42, U=0.604, f=0.494. Date: 2026-11-01*


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>
> Otsuki A. Comparing Genetic Algorithms and LLMs for Research-Hypothesis Generation: Mainstream Convergence of LLMs, Wider Hypothesis Exploration by Combinatorial Search, and the Limits of Literature-Distance Novelty Metrics. Informatics. 2026; 13(10):174. https://doi.org/10.3390/informatics13100174